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Joint modulation of 3-PPM and quantum squeezed states in communication systems

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***Abstract*—Pulse position modulation(PPM) is an excellent modulation form that is widely used for deep-space information transmission and also in quantum modulation form. Lots of related work proved that the squeezed state is superior to the traditional coherent state modulation in 3-PPM modulation. In this paper, a new modulation method is proposed, that is, two squeezed states are applied to 3-PPM modulation. Compared with one squeezed state modulation, it is found that the two squeezed state modulation has lower error probability.**

***Index Terms*—Pulse position modulation, Quantum modulation , Squeezed state, Error probability**

I. Introduction

In recent years, quantum theory has developed rapidly and has a promising future in many research fields. Quantum communication theory is produced by combining quantum theory with traditional communication theory. As a widely used traditional modulation mode, pulse position modulation(PPM) is adopted in free space optical transmission, and is a candidate for deep-space transmission [1] [2]. In this paper, quantum theory and traditional PPM modulation theory are combined to seek a better modulation mode.

Yuen et al analyzed quantum PPM systems in a famous paper [3]. They found the optimal elementary projectors using an algebraic method for this kind of modulation. In his work, G.Cariolaro proposed an original method for verifying geometrically uniform symmetry (GUS) based on square-root measurements (SRM) and quantum PPM properties, which is very intuitive and makes it possible to directly achieve the same optimal result [4]. G.Pierobon and G.Cariolaro deals with quantum PPM in their paper, both in the absence (pure states) and in the presence (mixed states) of thermal noise, using the Glauber representation of coherent laser radiation. In fact, SRM gives the minimum error probability for the GUS of the quantum PPM, which is the motivation for applying GUS to quantum PPM. Notably, they applied SRM and GUS to quantum PPM format, resulting in better results than classical PPM modulation, which is reflected on the error probability [5].

A feasible choice for PPM is to associate the bit 0 to the ground state $|\gamma_{ik}\rangle = |0, 0\rangle$ and to the bit 1 the generic state $|\gamma_{ik}\rangle = |z, \alpha\rangle$. This method has been proved to improve the performance of communication system [4]. On this basis, this paper proposes a different method, which is to associate the bit 0 to the state $|\gamma_{ik}\rangle = |z, \alpha\rangle$ and the bit 1 to the state $|\gamma_{ik}\rangle = |0, 0\rangle$. It is natural to wonder, since the squeezed state PPM modulation is superior to the coherent state modulation, whether multiple squeezed states acting together can achieve better performance. This paper proves that this idea is desirable.

II. Mathematical methods of quantum modulation and measurement

A. Classical PPM Format

In the classical format, the symbol period T is subdivided into K parts, and the duration of each part is $T_0 = T/K$. Then, to the symbol $i\epsilon\mathcal{A} = \{0, 1, ..., K-1\}$, its waveform consists of a series of rectangle pulses, mathematically expressed as follows:

$$\gamma_i(t) = \begin{cases} \Delta & iT_0 < t < (i+1)T_0 \\ 0 & elsewhere \end{cases} \quad i = 0, 1, \dots, K-1 \tag{1}$$

where $\Delta > 0$ is the scale factor. In this paper we will adopt the specular format

$$\gamma_i(t) = \begin{cases} \Delta & (K-1-i)T_0 < t < (K-i)T_0 \\ 0 & elsewhere \end{cases} \\ i = 0, 1, \dots, K-1 \tag{2}$$

where the ith position becomes $(K-1-i)T_0$ instead of iT_0. The reason for this is that this scheme greatly simplifies the form of symmetric operators [2].

To waveforms (2), K binary words can be associated of length K:

$$\gamma_i = [\gamma_{i,K-1}, \dots, \gamma_{i,1}, \gamma_{i,0}] \quad i = 0, 1, \dots, K-1 \tag{3}$$

where $\gamma_{ij} = \Delta\delta_{ij}$. For example, when K=3 the words are

$$\begin{aligned} \gamma_0 &= [0 \quad 0 \quad \Delta] \\ \gamma_1 &= [0 \quad \Delta \quad 0] \\ \gamma_2 &= [\Delta \quad 0 \quad 0] \end{aligned} \tag{4}$$

B. Quantum PPM Format

To classical PPM format, each word is associated with a displaced pulse in time domain, to quantum PPM format where the word is represented by a composite state. In other words, a composite quantum state replaces a time pulse. The coherent radiation of laser emission is modeled as coherent state. The set of coherent states is indicated by

$$\mathcal{G}=\{|\alpha\rangle, \alpha \epsilon \mathbb{C}\} \tag{5}$$

where α is a complex amplitude, whose meaning is that $|\alpha|^2$ is the average number of photons in state $|\alpha\rangle$. The tensor product of two or more coherent states will be particularly relevant to PPM modulation. This result can be represented as:

$$|\alpha\rangle=|\alpha_1\rangle \otimes|\alpha_2\rangle \otimes \ldots \otimes|\alpha_N\rangle. \tag{6}$$

We also know that the quantum formulation of PPM must be done over a composite Hilbert space, given by the tensor product $\mathcal{H}=\mathcal{H}_0 \otimes \mathcal{H}_0 \otimes \cdots \otimes \mathcal{H}_0$ of K equal Hilbert space $\mathcal{H}_0$ [6]. In the joint modulation of PPM and quantum states, the coherent states become:

$$|\gamma_i\rangle=|\gamma_{i, K-1}\rangle \otimes \cdots \otimes|\gamma_{i, 1}\rangle \otimes|\gamma_{i, 0}\rangle, \quad i=0,1, \ldots, K-1 \tag{7}$$

with

$$|\gamma_{ij}\rangle= \begin{cases}|\mathit{\Delta}\rangle & i=j \\ |0\rangle & i \neq j\end{cases} \tag{8}$$

where $|\mathit{\Delta}\rangle$ is a coherent state with parameter $\mathit{\Delta}$, and $|0\rangle$ is the "ground state". For example, for *K*=3 we have the three states:

$$\begin{aligned} & |\gamma_0\rangle=|0\rangle \otimes|0\rangle \otimes|\mathit{\Delta}\rangle \\ & |\gamma_1\rangle=|0\rangle \otimes|\mathit{\Delta}\rangle \otimes|0\rangle \\ & |\gamma_2\rangle=|\mathit{\Delta}\rangle \otimes|0\rangle \otimes|0\rangle\end{aligned} \tag{9}$$

C. Quantum Squeezed States

We know that a complex parameter α corresponds to a coherent state $|\alpha\rangle$. A squeezed state is often expressed as $|z, \alpha\rangle$, which is defined by two complex parameters, the displacement α and the squeeze factor $z=re^{i\theta}$. If we set z to zero, the squeezed-displaced state degenerates into a coherent state:

$$|0, \alpha\rangle=|\alpha\rangle \in \mathcal{G}. \tag{10}$$

This inspired us to use squeezed states in quantum communication system, which maybe get better performance. However, the premise of doing this is to choose the right parameters. The idea mentioned above have been proved to be correct [7] . In this paper, the idea will be further extended to improve the performance of quantum communication system.

Like coherent states, squeezed states are defined in an infinite Hilbert space. They may be specified by the Fock expansion, whose Fourier coefficients result in

$$|z, \alpha\rangle_n=\frac{\sqrt{n!}}{\mu}\left(\frac{\beta}{\mu}\right)^n \mathcal{H}_n\left(\frac{\mu \nu}{\beta^2}\right) \exp \left(-\frac{1}{2}|\beta|^2-\beta^2 \frac{\nu^*}{2 \mu}\right). \tag{11}$$

where $\mathcal{H}_n(x)$ are the polynomials

$$\mathcal{H}_n(x):=\sum_{j=0}^{[n/2]} \frac{1}{(n-2j)!j!} x^j. \tag{12}$$

and

$$\mu=\cosh r, \quad \nu=\sinh r \exp(i\theta), \quad \beta=\mu\alpha-\nu\alpha^*. \tag{13}$$

To evaluate the error probability of quantum communication systems with PPM, we need the following statistics of the single-mode state $|z, \alpha\rangle$:

- The mean photon number, which is given by [4]

$$\overline{N}_{|z, \alpha\rangle}=|\alpha|^2+\sinh^2(r) \tag{14}$$

  In equal symbol probability $(q_i=1/K)$, all the K PPM symbols have the same mean number of photons per symbol, given by

$$N_s=|\alpha|^2+\sinh^2(r) \tag{15}$$

- The inner product between two states was evaluated by Yuen[8]:

$$\begin{aligned}&\langle z_1, \alpha_1 | z_0, \alpha_0\rangle= \\ &A^{-\frac{1}{2}} exp\left[-\frac{AC-2\beta_1\beta_2^*+B\beta_1^{*2}-B^*\beta_0^2}{2A}\right]\end{aligned} \tag{16}$$

  where $\mu_i=\cosh(r_i), \nu_i=\sinh(r_i) e^{i\theta_i}, \beta_i=\mu_i\alpha_i-\nu_i\alpha_i^*, A=\mu_0\mu_1^*-\nu_0\nu_1^*, B=\nu_0\mu_1-\mu_0\nu_1, C=|\beta_1|^2+|\beta_0|^2$.

D. The Geometrically Uniform Symmetry(GUS)

Lots of related work proved SRM will give the minimum error probability for the GUS of the quantum PPM, which shows that GUS is critical to quantum PPM format.The context of GUS is provided by QC system where the transmission of classic information uses quantum states as physical carriers [9]. A classical source emits a symbol A belonging to a set of K elements, $A \in \mathcal{A}=\{0,1, \ldots, K-1\}$, with assigned a priori probabilities $q_i=P[A=i], i \in \mathcal{A}$. The transmitter (Alice) encodes the symbol A into a quantum state $|\gamma_A\rangle$ in a constellation of K pure states $\{|\gamma_0\rangle, \ldots,|\gamma_{K-1}\rangle\}$, and more generally, in a constellation of mixed states with density operators $\{\rho_0, \ldots, \rho_{K-1}\}$ [14].

When the following two conditions are met, a constellation of K pure states $\{|\gamma_0\rangle, \ldots,|\gamma_{K-1}\rangle\}$ has the GUS:

- there exists a unitary operator S with the property

$$|\gamma_i\rangle=S^i|\gamma_0\rangle, \quad i=0,1, \ldots, K-1 \tag{17}$$

- the operator S is a Kth root of the identity operator, the formula for that is

$$S^K=I_{\mathcal{H}^{\otimes N}} \tag{18}$$

  where $I_{\mathcal{H}^{\otimes N}}$ is the identity operator of $\mathcal{H}^{\otimes N}$.

In the presence of the GUS, state constellations are uniquely identified by the symmetry operator S and the reference state $|\gamma_0\rangle$.

The receiver (Bob) performs a quantum measurement from the received state ρ_A with POVM measurement operators $\{P_k, k \in \mathcal{A}\}$, which is a set of m operators $\prod_0, \ldots, \prod_{m-1}$ with $\prod_i = |\mu_i\rangle\langle\mu_i|$ and $|\mu_i\rangle = S^i|\mu_0\rangle$ [15]. Then we can get the correct decision probability:

$$P_c = \sum_{i \in \mathcal{A}} q_i Tr\left(\rho_i P_i\right) \tag{19}$$

For pure states, that is with $\rho_i = |\gamma_i\rangle\langle\gamma_i|$.

E. Square-Root Measurements(SRM)

In the absence of thermal noise(pure states), the measurement vectors $|\mu_i\rangle$ are chosen with the standard of minimizing the differences between the states and the measurement vectors [10], $|e_i\rangle = |\gamma_i\rangle - |\mu_i\rangle$, we look for the measurement vectors $|\mu_i\rangle$ which minimize the quadratic error

$$\varepsilon = \sum_{i=0}^{K-1} \langle e_i|e_i\rangle = \sum_{i=0}^{K-1} \left(\langle\gamma_i| - \langle\mu_i|\right)\left(|\gamma_i\rangle - |\mu_i\rangle\right) \tag{20}$$

with the constraint of the resolution of the identity $\sum_{i=0}^{K-1} |\mu_i\rangle\langle\mu_i| = I_{\mathcal{H}}$.

The evaluation of the measurement vector is obtained computing the inner product $\langle\gamma_i|\gamma_j\rangle$ between the states of the constellation [11], the measurement vectors result explicitly in $|\mu_i\rangle = \sum_{j=0}^{K-1}(G^{-\frac{1}{2}})_{ij}|\gamma_j\rangle$, thus obtaining the Gram's matrix:

$$G_{K\times K} = \Gamma^*\Gamma = \left[\langle\gamma_i|\gamma_j\rangle\right], \quad i,j = 0,\ldots,K-1 \tag{21}$$

The transition probabilities result in $p_c\left(j|i\right) = \left|(G^{1/2})_{ij}\right|^2$, from which we obtain the correct decision probability

$$P_c = \frac{1}{K}\sum_{i=0}^{K-1}\left|(G^{1/2})_{ii}\right|^2 \tag{22}$$

The advantage of the SRM method is that it gives optimal results for PPM constellation with GUS [15].

III. Application to 3-PPM Quantum

A. One Squeezed State in 3-PPM

In the quantum PPM, the quantum states live in a composite Hilbert space, which is given by the tensor product $\mathcal{H} = \mathcal{H}_0 \otimes \mathcal{H}_0 \otimes \cdots \otimes \mathcal{H}_0$ of K equal Hilbert space $\mathcal{H}_0$ [12], where $\mathcal{H}_0$ has *n* dimensions and $\mathcal{H}$ has $N = n^K$ dimensions:

$$|\gamma_i\rangle = |\gamma_{i,0}\rangle \otimes |\gamma_{i,1}\rangle \otimes \cdots \otimes |\gamma_{i,K-1}\rangle, \quad i = 0,1,\ldots,K-1 \tag{23}$$

The most general squeezed-displaced states is symbolized by $|z,\alpha\rangle$, one choice for PPM is to associate the bit 0 to the ground state $|\gamma_{ik}\rangle = |0,0\rangle$ and 1 to the generic state $|\gamma_{ik}\rangle = |z,\alpha\rangle$ [13]. For *K*=3 we have

$$\begin{aligned}
|\gamma_0\rangle &= |z,\alpha\rangle \otimes |0,0\rangle \otimes |0,0\rangle \\
|\gamma_1\rangle &= |0,0\rangle \otimes |z,\alpha\rangle \otimes |0,0\rangle \\
|\gamma_2\rangle &= |0,0\rangle \otimes |0,0\rangle \otimes |z,\alpha\rangle
\end{aligned} \tag{24}$$

In the SRM the error probability depends only on the inner product between the single-mode states $|z,\alpha\rangle$ and $|0,0\rangle$ [4]. In fact, the Gram matrix is given by

$$G = \begin{bmatrix} 1 & \Gamma & \ldots & \Gamma \\ \Gamma & 1 & \ldots & \Gamma \\ \vdots & \vdots & \ddots & \vdots \\ \Gamma & \Gamma & \ldots & 1 \end{bmatrix} \tag{25}$$

where $\Gamma := |\langle z,\alpha|0,0\rangle|^2$.

The squared inner product can be written in the form

$$\Gamma = \frac{1}{\cosh r} exp\left[-\alpha^2 f\left(r,\theta\right)\right] \tag{26}$$

where

$$\begin{aligned} f\left(r,\theta\right) = &\cosh(2r) + \tanh(r)\sinh(2r) - \\ &[\sinh(2r) + \tanh(r)\cosh(2r)]\cos(\theta) \end{aligned} \tag{27}$$

Then the error probability becomes

$$P_e = 1 - \frac{1}{K^2}\left(\sqrt{1+(K-1)\Gamma} + (K-1)\sqrt{1-\Gamma}\right)^2 \tag{28}$$

Also in this case P_e can be expressed in terms of the mean photon number per symbol N_s, by writing Γ as

$$\Gamma = \frac{1}{\cosh r} exp\left[-(N_s - \sinh^2 r) f\left(r,\theta\right)\right] \tag{29}$$

Although Cariolaro G et al have given the optimal value of θ in 4-PPM modulation format , it is still necessary to explore the best value of θ in 3-PPM modulation format [4]. In order to explore the influence of phase θ on error probability in squeezed 3-PPM modulation format, r can be kept in constants to simulate the system. The simulation results are as follows:

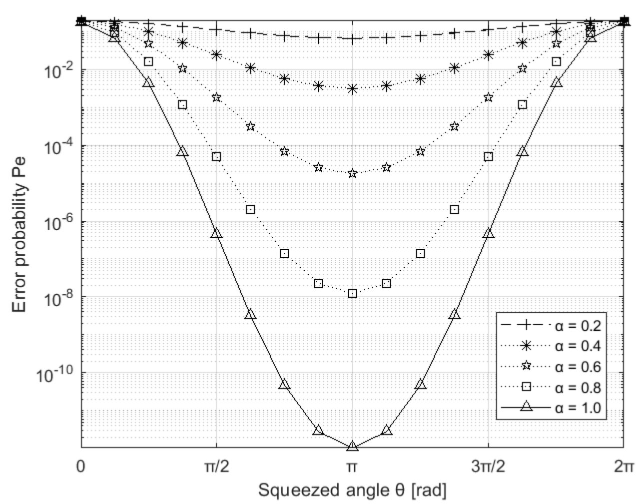

Fig. 1. Error probability P_e versus the phase θ for r=1.0 and five values of the displacement

From the Fig.1 we can easily realize that a squeezing factor with $\theta = \pi$ gives the best performance. At this point, it is not difficult to get the constellation diagram of joint modulation of 3-PPM and quantum squeezed states, as shown in Fig.2.

What needs illustration is that the ellipses around the states represent the contour levels of the Wigner function of each squeezed states [4], and the eccentricity of the ellipse only depends on the squeezed factor r.

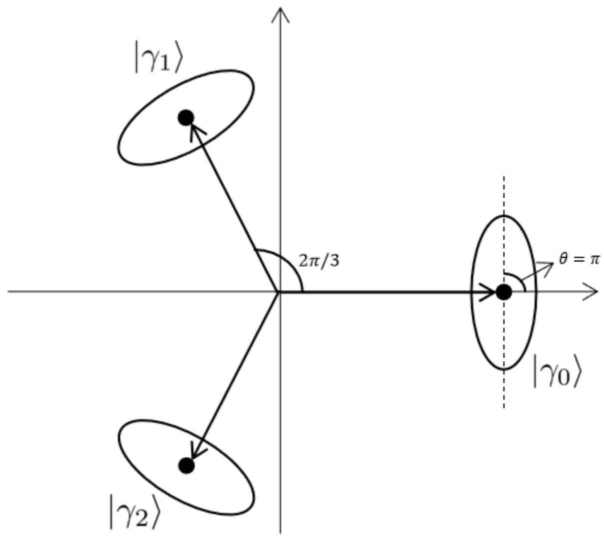

Fig. 2. Constellation of 3-PPM quantum squeezed states with the GUS in the complex plane $\mathbb{C}$.

In Fig. 3, we compare the Error probability of the coherent state with that of the squeezed states at different r, which proves that when the average photon number is small(weak signal), the squeezed state is superior to the coherent state.

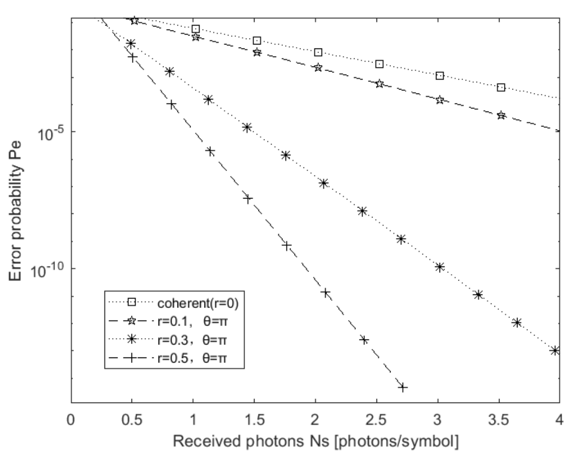

Fig. 3. Error probability of quantum 3-PPM for coherent states and three values of squeezing.

B. Two Squeezed States in 3-PPM

On the basis of the above design, this paper proposes another modulation method, which associate the bit 0 to the generic state $|\gamma_{ik} = |z, \alpha\rangle$ and the bit 1 to the ground state $|\gamma_{ik} = |0, 0\rangle$.For K=3, we have

$$\begin{aligned} |\gamma_0\rangle &= |0,0\rangle \otimes |z_0,\alpha_0\rangle \otimes |z_1,\alpha_1\rangle \\ |\gamma_1\rangle &= |z_0,\alpha_0\rangle \otimes |z_1,\alpha_1\rangle \otimes |0,0\rangle \\ |\gamma_2\rangle &= |z_1,\alpha_1\rangle \otimes |0,0\rangle \otimes |z_0,\alpha_0\rangle \end{aligned} \tag{30}$$

In this case, we also let a generic complex squeezing factor $z = re^{i\theta}$. The difference is that there are now two squeezed states: $z_0 = r_0 e^{i\theta_0}$ and $z_1 = r_1 e^{i\theta_1}$. In 3-PPM format, the Gram matrix is

$$G = \begin{bmatrix} 1 & \Gamma & \Gamma \\ \Gamma & 1 & \Gamma \\ \Gamma & \Gamma & 1 \end{bmatrix} \tag{31}$$

And not only that, the inner product Γ is no longer the square inner product. Instead, it becomes something like this:

$$\begin{aligned} \Gamma &= \langle z_0,\alpha_0|z_1,\alpha_1\rangle\langle z_1,\alpha_1|0,0\rangle\langle 0,0|z_0,\alpha_0\rangle \\ &= A^{-\frac{1}{2}} exp\left[-\frac{AC - 2\beta_1\beta_0^* + B\beta_1^{*2} - B^*\beta_0^2}{2A}\right]\cdot \\ &\sqrt{\frac{1}{\cosh r_1} exp\left[-\alpha_1^2 f(r_1,\theta_1)\right]}\cdot \\ &\sqrt{\frac{1}{\cosh r_0} exp\left[-\alpha_0^2 f(r_0,\theta_0)\right]} \end{aligned} \tag{32}$$

where $\mu_i = \cosh(r_i), \nu_i = \sinh(r_i) e^{i\theta_i}, \beta_i = \mu_i\alpha_i - \nu_i\alpha_i^*, A = \mu_0\mu_1^* - \nu_0\nu_1^*, B = \nu_0\mu_1 - \mu_0\nu_1, C = |\beta_1|^2 + |\beta_0|^2$

Obviously, there are two independent squeezed states that affect the modulation performance. Each squeezed state has an independent number of photons. We investigate the effect of two independent squeezed states on error probability Pe as Fig.4.

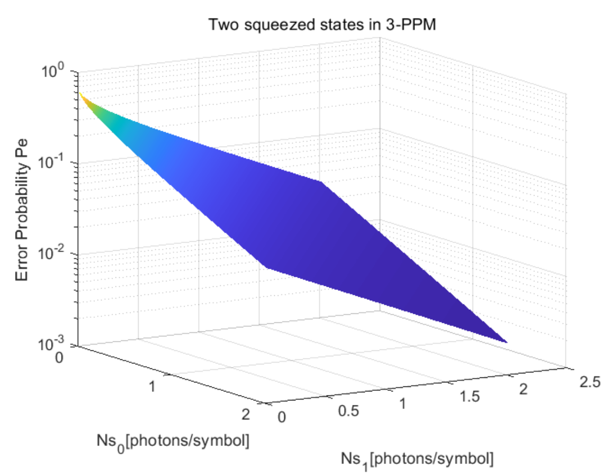

Fig. 4. Error probability of quantum 3-PPM for two squeezed states and two values of photon numbers per symble.

In order to compare the performance of this system with that of an squeezed state, we can make the photon number of an squeezed state a constant, and then compare the performance of the two modulation formats. From the discussion above, we know that the best θ is π. We assume that both squeezed states are at the optimum θ, i.e.$\theta_0 = \theta_1 = \pi$, and $r = 0.1$. The simulation results of the system are as Fig.5.

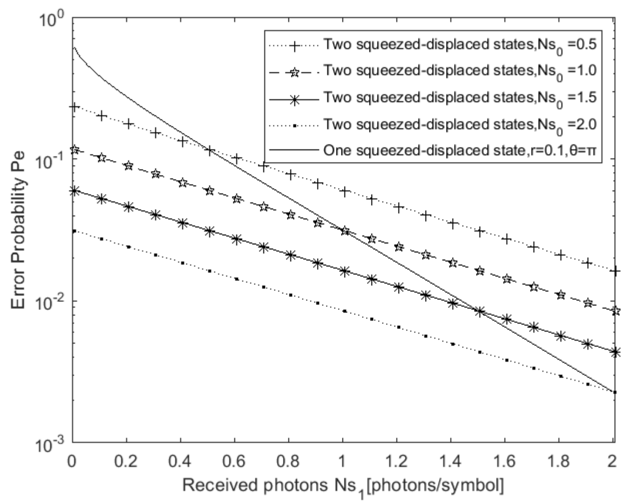

Fig. 5. Error probability of quantum 3-PPM for two squeezed states versus different Ns_1.

By comparison, we find that the joint modulation of two squeezed states and 3-PPM has better performance than 3-PPM in one squeezed state. And with the increase of photon number, Pe decreases continuously.

IV. Conclusion

In the case of joint modulation of 3-PPM and quantum squeezed states in communication systems, its performance is proved to be superior to coherent state modulation when the average photon number is small(weak signal). The application of two squeezed states in the 3-PPM modulation system is proved to be superior to that of one squeezed state. And the optimum $\theta = \pi$. The error probability Pe decreases with the increase of squeezing factor r and photon numbers per symbol Ns .

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